Lab Exercise 13- Managing Namespaces in Kubernetes

Step 1: Understand Namespaces

Namespaces provide a mechanism for scoping resources in a cluster. Namespaces can be used to:

- Create environments for different applications or teams.
- Apply policies like resource quotas or network policies on a per-namespace basis.
- Separate operational environments (like development and production).

Step 2: List Existing Namespaces

To list all the namespaces in your Kubernetes cluster:

```
kubectl get namespaces
```

You will typically see default namespaces like default, kube-system, and kube-public.

Step 3: Create a Namespace

You can create a namespace using a YAML file or directly with the kubectl command.

Using YAML File

Create a file named my-namespace.yaml with the following content:

```
apiVersion: v1
kind: Namespace
metadata:
  name: my-namespace
```

Apply this YAML to create the namespace:

```
kubectl apply -f my-namespace.yaml
```

Using kubectl Command

Alternatively, create a namespace using the kubectl command:

```
kubectl create namespace my-namespace
```

Verify that the namespace is created:

```
kubectl get namespaces
```

You should see my-namespace listed in the output.

Step 4: Deploy Resources in a Namespace

Create resources such as Pods, Services, or Deployments within the new namespace.

Deploy a Pod in the Namespace

Create a YAML file named nginx-pod.yaml with the following content:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  namespace: my-namespace
spec:
  containers:
  - name: nginx
    image: nginx:latest
    ports:
    - containerPort: 80
```

Apply this YAML to create the Pod:

```
kubectl apply -f nginx-pod.yaml
```

Check the status of the Pod within the namespace:

```
kubectl get pods -n my-namespace
```

To describe the Pod and see detailed information:

```
kubectl describe pod nginx-pod -n my-namespace
```

Create a Service in the Namespace

Create a YAML file named **nginx-service.yaml** with the following content:

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
  namespace: my-namespace
spec:
  selector:
    app: nginx-pod
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: ClusterIP
```

Apply this YAML to create the Service:

```
kubectl apply -f nginx-service.yaml
```

Check the status of the Service within the namespace:

```
kubectl get services -n my-namespace
```

To describe the Service and see detailed information:

```
kubectl describe service nginx-service -n my-namespace
```

Step 5: Switching Context Between Namespaces

When working with multiple namespaces, you can specify the namespace in kubectl commands or switch the default context.

Specify Namespace in Commands

You can specify the namespace directly in kubectl commands using the -n or --namespace flag:

```
kubectl get pods -n my-namespace
```

Set Default Namespace for kubectl Commands

To avoid specifying the namespace every time, you can set the default namespace for the current context:

```
kubectl config set-context --current --namespace=my-namespace
```

Verify the current context's namespace:

```
kubectl config view --minify | grep namespace
```

Step 6: Clean Up Resources

To delete the resources and the namespace you created:

```
kubectl delete -f nginx-pod.yaml
kubectl delete -f nginx-service.yaml
kubectl delete namespace my-namespace
```

Ensure that the namespace and all its resources are deleted:

```
kubectl get namespaces
```

```
C:\Users\NK>kubectl get namespaces
NAME                   STATUS   AGE
default                Active   47m
kube-node-lease        Active   47m
kube-public            Active   47m
kube-system            Active   47m
kubernetes-dashboard   Active   33m

C:\Users\NK>(echo apiVersion: v1 & echo kind: Namespace & echo metadata: & echo   name: my-namespace) > my-namespace.yaml

C:\Users\NK>kubectl apply -f my-namespace.yaml
namespace/my-namespace created

C:\Users\NK>kubectl get namespaces
NAME                   STATUS   AGE
default                Active   47m
kube-node-lease        Active   47m
kube-public            Active   47m
kube-system            Active   47m
kubernetes-dashboard   Active   33m
my-namespace           Active   8s

C:\Users\NK>(echo apiVersion: v1 & echo kind: Pod & echo metadata: & echo   name: nginx-pod & echo   namespace: my-namespace & echo spec: & echo   container
s: & echo   - name: nginx & echo     image: nginx:latest & echo     ports: & echo     - containerPort: 80) > nginx-pod.yaml

C:\Users\NK>kubectl apply -f nginx-pod.yaml
pod/nginx-pod created

C:\Users\NK>kubectl get pods -n my-namespace
NAME        READY   STATUS    RESTARTS   AGE
nginx-pod   1/1     Running   0          8s

C:\Users\NK>kubectl describe pod nginx-pod -n my-namespace
Name:             nginx-pod
Namespace:        my-namespace
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
```

```
nginx-pod   1/1     Running   0          8s

C:\Users\NK>kubectl describe pod nginx-pod -n my-namespace
Name:             nginx-pod
Namespace:        my-namespace
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 28 Apr 2026 09:33:44 +0530
Labels:           <none>
Annotations:      <none>
Status:           Running
IP:               10.244.0.51
IPs:
  IP:  10.244.0.51
Containers:
  nginx:
    Container ID:   docker://d0e55c18092777f0de6852dae0d5ce6e4e353832ad7b70dde0d566b39aae4012
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Tue, 28 Apr 2026 09:33:48 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-zst2r (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-zst2r:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
```

```
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  16s   default-scheduler  Successfully assigned my-namespace/nginx-pod to minikube
  Normal  Pulling    15s   kubelet            Pulling image "nginx:latest"
  Normal  Pulled     12s   kubelet            Successfully pulled image "nginx:latest" in 2.533s (2.533s including waiting). Image size: 160939956 bytes.
  Normal  Created    12s   kubelet            Container created
  Normal  Started    12s   kubelet            Container started

C:\Users\NK>(echo apiVersion: v1 & echo kind: Service & echo metadata: & echo   name: nginx-service & echo   namespace: my-namespace & echo spec: & echo   s
elector: & echo     app: nginx-pod & echo   ports: & echo   - protocol: TCP & echo     port: 80 & echo     targetPort: 80 & echo   type: ClusterIP) > nginx-
service.yaml

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service.yaml

C:\Users\NK>kubectl apply -f nginx-service.yaml
service/nginx-service created

C:\Users\NK>kubectl get services -n my-namespace
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
nginx-service   ClusterIP   10.96.83.206   <none>        80/TCP    2s

C:\Users\NK>kubectl describe service nginx-service -n my-namespace
Name:                     nginx-service
Namespace:                my-namespace
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx-pod
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.96.83.206
IPs:                      10.96.83.206
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:
Session Affinity:         None
```


```
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
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  Normal  Scheduled  16s   default-scheduler  Successfully assigned my-namespace/nginx-pod to minikube
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NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
nginx-service   ClusterIP   10.96.83.206   <none>        80/TCP    2s

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Name:                     nginx-service
Namespace:                my-namespace
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx-pod
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.96.83.206
IPs:                      10.96.83.206
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:
Session Affinity:         None
```